

An Analysis of ResNet deep learning

ResNet deep learning

$$y = F(x, \{W_i\}) + x$$

ResNet deep learning -- Part 1

Mobile advertising Finding the appropriate mobile audience for mobile advertising is always challenging, since many data points must be considered and analyzed before a target segment can be created and used in ad serving by any ad server. Deep learning has been used to interpret large, many-dimensioned advertising datasets. Many data points are collected during the request/serve/click internet advertising cycle. This information can form the basis of machine learning to improve ad selection.

ResNet deep learning -- Part 2

Deep neural networks A deep neural network (DNN) is an artificial neural network with multiple layers between the input and output layers. There are different types of neural networks but they always consist of the same components: neurons, synapses, weights, biases, and functions. These components as a whole function in a way that mimics functions of the human brain, and can be trained like any other ML algorithm. For example, a DNN that is trained to recognize dog breeds will go over the given image and calculate the probability that the dog in the image is a certain breed. The user can review the results and select which probabilities the network should display (above a certain threshold, etc.) and return the proposed label. Each mathematical manipulation as such is considered a layer, and complex DNN have many layers, hence the name "deep" networks. DNNs can model complex non-linear relationships. DNN architectures generate compositional models where the object is expressed as a layered composition of primitives. The extra layers enable composition of features from lower layers, potentially modeling complex data with fewer units than a similarly performing shallow network. For instance, it was proved that sparse multivariate polynomials are exponentially easier to approximate with DNNs than with shallow networks. Deep architectures include many variants of a few basic approaches. Each architecture has found success in specific domains. It is not always possible to compare the performance of multiple architectures, unless they have been evaluated on the same data sets. DNNs are typically feedforward networks in which data flows from the input layer to the output layer without looping back. At first, the DNN creates a map of virtual neurons and assigns random numerical values, or "weights", to connections between them. The weights and inputs are multiplied and return an output between 0 and 1. If the network did not accurately recognize a particular pattern, an algorithm would

adjust the weights. That way the algorithm can make certain parameters more influential, until it determines the correct mathematical manipulation to fully process the data. Recurrent neural networks, in which data can flow in any direction, are used for applications such as language modeling. Long short-term memory is particularly effective for this use. Convolutional neural networks (CNNs) are used in computer vision. CNNs also have been applied to acoustic modeling for automatic speech recognition (ASR).

ResNet deep learning -- Part 3

Deep backward stochastic differential equation method Deep backward stochastic differential equation method is a numerical method that combines deep learning with Backward stochastic differential equation (BSDE). This method is particularly useful for solving high-dimensional problems in financial mathematics. By leveraging the powerful function approximation capabilities of deep neural networks, deep BSDE addresses the computational challenges faced by traditional numerical methods in high-dimensional settings. Specifically, traditional methods like finite difference methods or Monte Carlo simulations often struggle with the curse of dimensionality, where computational cost increases exponentially with the number of dimensions. Deep BSDE methods, however, employ deep neural networks to approximate solutions of high-dimensional partial differential equations (PDEs), effectively reducing the computational burden. In addition, the integration of Physics-informed neural networks (PINNs) into the deep BSDE framework enhances its capability by embedding the underlying physical laws directly into the neural network architecture. This ensures that the solutions not only fit the data but also adhere to the governing stochastic differential equations. PINNs leverage the power of deep learning while respecting the constraints imposed by the physical models, resulting in more accurate and reliable solutions for financial mathematics problems.

ResNet deep learning -- Part 4

Relation to human cognitive and brain development Deep learning is closely related to a class of theories of brain development (specifically, neocortical development) proposed by cognitive neuroscientists in the early 1990s. These developmental theories were instantiated in computational models, making them predecessors of deep learning systems. These developmental models share the property that various proposed learning dynamics in the brain (e.g., a wave of nerve growth factor) support the self-organization somewhat analogous to the neural networks utilized in deep learning models. Like the neocortex, neural networks employ a hierarchy of layered filters

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in which each layer considers information from a prior layer (or the operating environment), and then passes its output (and possibly the original input), to other layers. This process yields a self-organizing stack of transducers, well-tuned to their operating environment. A 1995 description stated, "...the infant's brain seems to organize itself under the influence of waves of so-called trophic-factors ... different regions of the brain become connected sequentially, with one layer of tissue maturing before another and so on until the whole brain is mature". A variety of approaches have been used to investigate the plausibility of deep learning models from a neurobiological perspective. On the one hand, several variants of the backpropagation algorithm have been proposed in order to increase its processing realism. Other researchers have argued that unsupervised forms of deep learning, such as those based on hierarchical generative models and deep belief networks, may be closer to biological reality. In this respect, generative neural network models have been related to neurobiological evidence about sampling-based processing in the cerebral cortex. Although a systematic comparison between the human brain organization and the neuronal encoding in deep networks has not yet been established, several analogies have been reported. For example, the computations performed by deep learning units could be similar to those of actual neurons and neural populations. Similarly, the representations developed by deep learning models are similar to those measured in the primate visual system both at the single-unit and at the population levels.

ResNet deep learning -- Part 5

A large percentage of candidate drugs fail to win regulatory approval. These failures are caused by insufficient efficacy (on-target effect), undesired interactions (off-target effects), or unanticipated toxic effects. Research has explored use of deep learning to predict the biomolecular targets, off-targets, and toxic effects of environmental chemicals in nutrients, household products and drugs. AtomNet is a deep learning system for structure-based rational drug design. AtomNet was used to predict novel candidate biomolecules for disease targets such as the Ebola virus and multiple sclerosis. In 2017 graph neural networks were used for the first time to predict various properties of molecules in a large toxicology data set. In 2019, generative neural networks were used to produce molecules that were validated experimentally all the way into mice.